

PORVOO KALLBADAGRUND, Finland

WMO: 029870

Lat: 59.9853N

Lon: 25.5986E

Elev: 23

StdP: 101.05

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.9	-12.1	-18.6	0.7	-12.9	-16.1	0.9	-10.8	18.7	-9.6	17.2	-4.0	6.4	50	0.892

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.0	23.1	19.8	21.8	18.9	20.5	17.7	20.5	22.4	19.2	21.2	18.0	19.8	5.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.7	14.5	22.0	18.4	13.3	20.6	17.2	12.3	19.4	59.2	22.5	54.9	21.2	51.0	19.7	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.6	15.9	14.4	DB	-15.3	25.0	3.9	1.7	-18.2	26.3	-20.5	27.3	-22.7	28.2	-25.5	29.5	(4)
(5)			WB	-16.0	20.6	3.8	1.2	-18.7	21.5	-21.0	22.2	-23.1	22.9	-25.9	23.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-2.5	-3.4	-0.7	3.4	9.2	13.3	17.5	17.2	13.5	8.0	3.9	0.9		
	DBStd	7.87	4.44	4.89	3.26	2.37	3.40	2.54	2.68	2.29	2.40	3.04	3.33	3.77		
	HDD10.0	1894	387	374	330	198	56	3	0	0	4	76	183	282		
	HDD18.3	4268	645	608	588	448	284	153	51	49	146	322	433	541		
	CDD10.0	708	0	0	0	0	31	100	231	224	109	13	0	0		
	CDD18.3	39	0	0	0	0	1	1	23	15	0	0	0	0		
	CDH23.3	31	0	0	0	0	1	1	23	7	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	7.8	8.8	7.8	7.8	7.1	6.9	7.0	6.4	6.7	7.4	8.7	9.4	9.9		
(15) Precipitation	PrecAvg	854	87	66	56	41	53	80	75	88	62	82	82	84		
	PrecMax	1217	181	143	113	102	100	170	163	202	110	140	132	142		
	PrecMin	452	25	15	16	2	10	24	17	13	18	10	14	37		
	PrecStd	172	37	33	27	22	26	36	35	42	25	30	33	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	4.1	8.0	11.7	20.7	21.3	25.4	24.3	19.1	14.2	10.2	8.2		
		MCWB	4.5	2.0	4.1	7.3	12.6	15.9	20.1	20.7	16.8	12.8	9.3	7.3		
	2%	DB	4.7	3.1	5.0	9.5	17.5	19.4	23.7	22.6	17.7	13.5	9.5	7.2		
		MCWB	3.7	1.8	1.9	6.2	11.9	15.4	20.4	19.5	15.8	12.3	8.6	6.1		
	5%	DB	3.9	2.4	3.7	7.7	15.4	17.9	22.6	21.3	17.0	12.8	8.7	6.2		
		MCWB	3.0	1.4	1.4	5.2	11.5	14.4	19.7	18.3	15.1	11.6	7.7	5.1		
	10%	DB	3.0	1.8	2.7	6.4	13.6	16.6	21.4	20.2	16.4	12.0	7.9	5.3		
		MCWB	2.1	0.8	1.0	4.3	10.4	13.9	18.8	17.4	14.6	10.8	6.8	4.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	3.1	4.6	8.1	14.7	18.1	21.7	21.2	17.3	13.2	9.6	7.5		
		MCDB	5.4	3.5	7.1	10.4	17.1	19.9	23.6	23.0	18.2	13.9	10.1	8.0		
	2%	WB	3.9	2.3	3.1	6.7	13.1	16.6	21.0	19.6	16.4	12.6	8.8	6.3		
		MCDB	4.4	2.7	4.1	8.2	15.5	18.5	22.9	21.8	17.4	13.3	9.3	7.0		
	5%	WB	3.1	1.7	2.3	5.4	12.0	15.3	20.1	18.5	15.7	11.9	7.9	5.3		
		MCDB	3.7	2.2	3.2	6.8	14.7	17.1	22.0	20.4	16.8	12.7	8.6	6.1		
	10%	WB	2.3	1.2	1.7	4.5	10.8	14.3	19.0	17.7	14.9	11.0	7.1	4.3		
		MCDB	3.0	1.7	2.4	5.9	13.4	16.1	20.9	19.3	16.2	11.9	7.7	5.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.3	3.0	3.0	3.6	4.3	3.5	3.0	2.7	2.6	2.7	2.6	3.0		
		MCDBR	3.0	2.5	3.8	5.7	6.7	5.3	4.0	3.6	2.8	2.7	2.7	2.9		
	5% WB	MCWBR	3.0	2.3	2.4	3.4	3.7	3.2	2.9	2.7	2.5	2.8	2.9	3.0		
		MCDWR	3.0	2.5	3.4	5.3	6.0	4.4	3.4	3.3	2.6	2.7	2.6	2.9		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.310	0.332	0.364	0.362	0.360	0.382	0.376	0.349	0.334	0.308	0.265		
		taud	2.149	2.182	2.223	2.307	2.410	2.478	2.439	2.452	2.512	2.399	2.220	2.104		
	Ebn at Noon		483	675	789	834	865	873	841	816	782	658	443	346		
		Edh at Noon	43	73	96	107	104	99	101	92	74	60	41	30		
(44) All-Sky Solar Radiation	RadAvg		0.25	0.82	2.31	3.86	5.31	5.52	5.33	4.14	2.52	1.09	0.30	0.15		
	RadStd		0.04	0.15	0.30	0.41	0.57	0.42	0.48	0.42	0.32	0.16	0.05	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	-146	N/A	N/A	N/A

Nomenclature: See separate page